

## Characterization of extended-spectrum cephalosporin-resistant *Salmonella enterica* serovar Heidelberg isolated from food animals, retail meat, and h...



*Salmonella enterica* is one of the most common causes of foodborne illness in the United States. Although salmonellosis is usually self-limiting, severe infections typically require antimicrobial treatment, and ceftriaxone, an extended-spectrum cephalosporin (ESC), is commonly used in both adults and children. Surveillance conducted by the National Antimicrobial Resistance Monitoring System (NARMS) has shown a recent increase in ESC resistance among *Salmonella* Heidelberg isolated from food animals at slaughter, retail meat, and humans. ESC resistance among *Salmonella* in the United States is usually mediated by a plasmid-encoded bla(CMY)  $\beta$ -lactamase. In 2009, we identified 47 ESC-resistant bla(CMY)-positive Heidelberg isolates from humans (n=18), food animals at slaughter (n=16), and retail meats (n=13) associated with a spike in the prevalence of this serovar. Almost 90% (26/29) of the animal and meat isolates were isolated from chicken carcasses or retail chicken meat. We screened NARMS isolates for the presence of bla(CMY), determined whether the gene was plasmid-encoded, examined pulsed-field gel electrophoresis patterns to assess the genetic diversities of the isolates, and categorized the bla(CMY) plasmids by plasmid incompatibility groups and plasmid multi-locus sequence typing (pMLST). All 47 bla(CMY) genes were found to be plasmid encoded. Incompatibility/replicon typing demonstrated that 41 were IncI1 plasmids, 40 of which only conferred bla(CMY)-associated resistance. Six were IncA/C plasmids that carried additional resistance genes. pMLST of the IncI1-bla(CMY) plasmids showed that 27 (65.8%) were sequence type (ST) 12, the most common ST among bla(CMY)-IncI1 plasmids from Heidelberg isolated from humans. Ten plasmids had a new ST profile, ST66, a type very similar to ST12. This work showed that the 2009 increase in ESC resistance among *Salmonella* Heidelberg was caused mainly by the dissemination of bla(CMY) on IncI1 and IncA/C plasmids in a variety of genetic backgrounds, and is likely not the result of clonal expansion.